

MS1: Concept Introduction

Peijing Xu, Matt Hoover, Ben Driscoll, Miles Lafferty, David Rowe



Rensselaer₂₀₀

Agenda

1. The Problem Statement
2. Targeted Customers
3. Customer Prioritization
4. Product Specification
5. Ergonomics and Safety Concerns
6. Benchmarking
7. What are the Benefits
8. Concept Design
9. Key Subsystems
10. Plans to Test Performance Specification

Our Customer

A kindergartener with an ataxic gait pattern

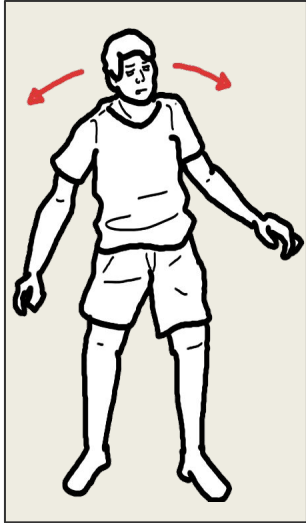
- Has a challenge with moving, motor control, and speech
- Currently in a special education classroom
 - 1 on 1 assistance for mobility
 - Classmates predominantly have mental disabilities
 - Wants to advance to standard class pathway
- Crawls on the belly to move independently
 - Can push a wheelchair (slowly)
 - Working to push a walker slowly
- Needs assistance getting into the standing position and back to sitting
- He is in the process of getting a full-time wheelchair

Customer needs assistance in his independent mobility in a new classroom

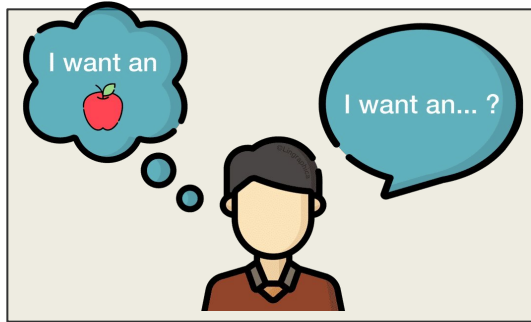


What is our Problem?

Ataxic Gait Pattern



Aphasia



Accessible School Environment for

- **Ataxic Gait Pattern**
 - Uncoordinated, unstable, and wide-based walking pattern
 - Twitchy lateral movements
- **Aphasia**
 - Language disorder that makes it difficult to understand or express language, read, and write.

Customer Needs (Prioritized)

Priority (1 is most important):

1. Safety
2. Assist user in moving from wheelchair to walker
3. Ease of transportation and movement
4. Strengthen / Exercise upper body
5. Hold objects steadily
6. Communication aid
7. Ease of language



Specifications

Customer Needs	Technical and Performance Specifications	
	Metric	Target Values
User Comfort	Weight (kg)	8-12
Size	Height (cm)	60
Stability	Max weight (kg)	20
Durability	Force (N)	45-75

Ergonomics and Safety Concerns



Product should be able to help transition target customers from Sitting to Standing

- Product must not break
- Product must not exhaust the customer
- Product must safely prevent customers from losing their balance like falling down or falling over

Benchmarking

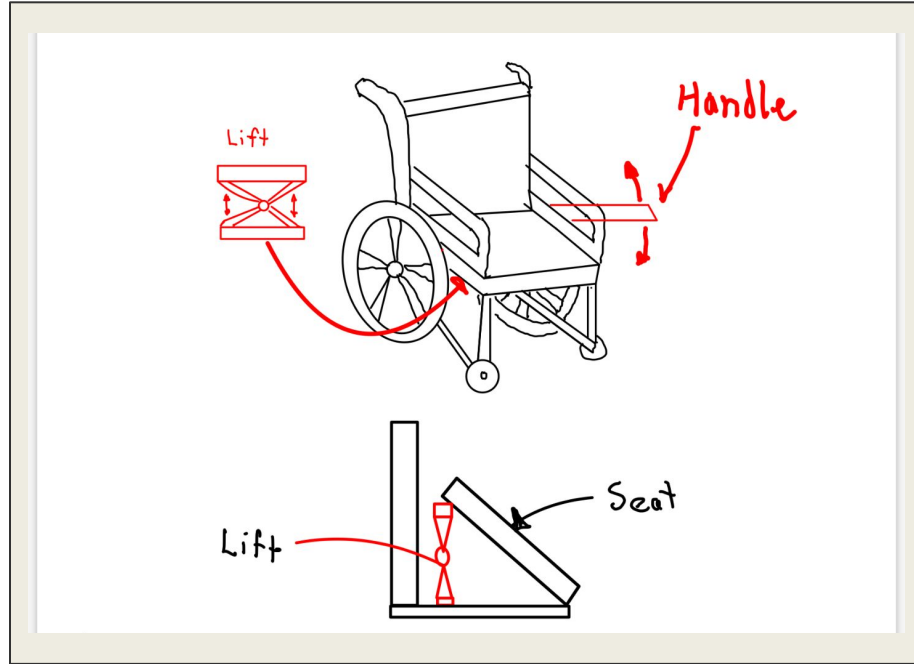


Overall Concept Performance

As we gather more specific information about the customer we hope to develop a device that:

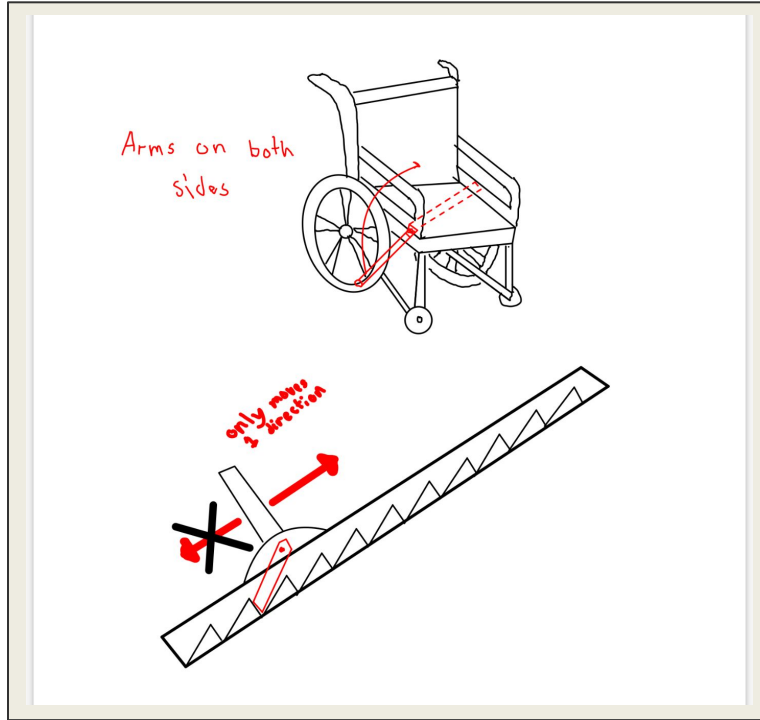
1. Is robust and safe
2. Converts the user's upper body strength into position adjustment
 - From seated to standing stable with walker
3. Will assist in stabilizing him in his lateral movements in this process

Concept 1



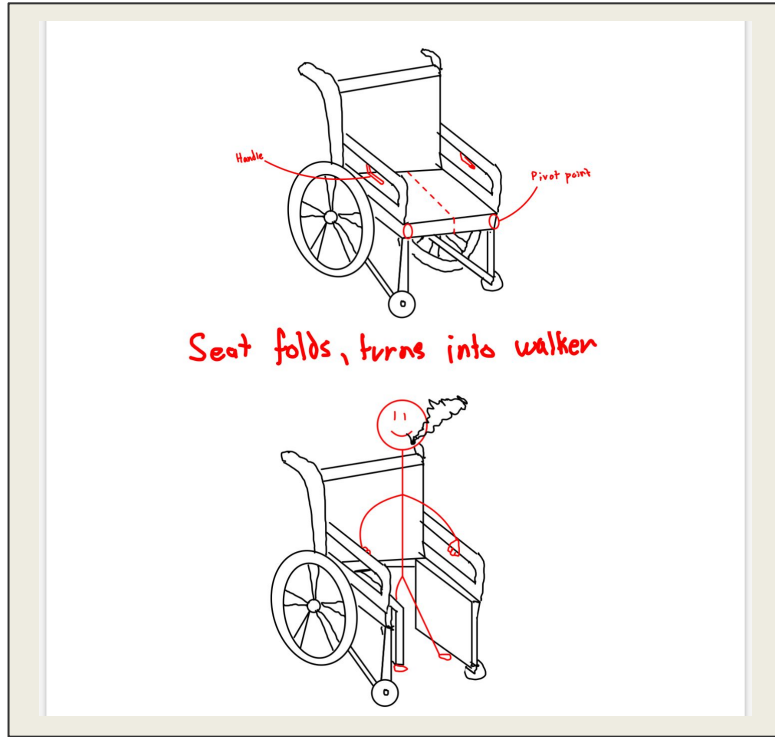
“Manual Scissor lift”
Lifts seat from the
back raising from
seated to standing
position

Concept 2



“Ratchet systems”
Can move handles
up but can't move
them back down

Concept 3



“Foldable Seat”
Chair essentially
turns into a walker

Benefits if Successful

Benefits:

- Increased Independence / confidence
- Allows for customer to advance into more challenging classes
- Less strain on user
- Space efficiency
- Lower need of caregivers
- Encouragement to using walker



Subsystems

- Overall Structure
 - Linking other subsystems to existing wheelchair and walker
- Braking System
 - Neither wheelchair nor walker can move while user is standing
- Lifting System
 - Depending on the concept, this could be different
- Stabilizing System
 - It may be necessary for the user's body to be supported during the process
- Sitting system
 - User must be able to return to seated position

Conclusion

- Our product provides **safety, independence**, and better **mobility**.
- It supports **seated-to-standing transitions** and strengthens **upper body**.
- Focus on **stability, ease of movement**, and **ergonomics** ensures effectiveness.
- Our main goal is reducing caregiver dependency and helping the customer advance.

Thank you!

Sources

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